

Crisis Logistics AI Optimizer – Live Demonstration Guide

Core Algorithm Demonstrations

DEMO 1: Max-Heap Priority Ranking

Objective

Demonstrate how incoming disaster relief requests are prioritized using a Binary Max-Heap.

Steps

1. Open the **Requests & Heap** tab.
2. Under **Generate Crisis Relief Request**, submit:

Request A

- Zone: Zone E (Santa Monica)
- Severity: Low
- Resource: Medicine
- Quantity: 50

Request B

- Zone: Zone A (Glendale)
- Severity: Critical
- Resource: Water Crate
- Quantity: 200

3. Observe the **Priority Heap Structure** panel.

Explanation

- The system automatically calculates a priority score for each request.
- Request B receives a higher score due to its critical severity.
- It automatically moves to the root of the heap.
- Request A remains at a lower level.

Mathematical Model

Priority Score = (Severity Weight × 10) + (Population ÷ 10,000)

Complexity Analysis

Insertion: $O(\log n)$

Deletion: $O(\log n)$

This ensures rapid prioritization even when thousands of requests are received.

DEMO 2: Greedy Resource Allocation

Objective

Demonstrate resource allocation based on request priority.

Steps

1. Click **Allocate Resources**.
2. Observe the request queue.

Possible statuses:

- Allocated
- Partial Allocation
- Unfulfilled

3. Navigate to the **Resource Inventory** tab.

Explanation

- The system always serves the highest-priority request first.
- Inventory updates instantly after allocation.
- If inventory is insufficient, a partial allocation is performed.

Algorithm Used

Greedy Allocation

Allocation = $\min(\text{Requested Quantity}, \text{Available Quantity})$

Benefit

Maximizes immediate relief to the most critical zones.

DEMO 3: Risk-Weighted Dijkstra Routing

Objective

Demonstrate how the system chooses the safest route rather than merely the shortest route.

Steps

1. Open **Route Visualizer**.
2. Select Zone H (Malibu) or Zone J (Anaheim).
3. Click **Inspect Route**.

Observation

- The route is highlighted on the map.
- Flooded and blocked roads are avoided.
- The safest route is displayed in the sidebar.

Mathematical Model

Effective Route Cost

= Distance $\times$ Hazard Multiplier

Road Types:

Normal Road = 1.0

Flooded Road = 3.0

Blocked Road = 9999.0

Algorithm Used

Dijkstra's Algorithm

Time Complexity:

$O(E \log V)$

Where:

V = Number of Nodes

E = Number of Roads

DEMO 4: Knapsack Vehicle Cargo Optimization

Objective

Demonstrate optimal vehicle loading under limited capacity.

Steps

1. Select a crisis scenario.
2. Click **Run Simulation**.
3. Open the **Fleet Assignment** tab.
4. Scroll to **Cargo Load Optimization**.

Observation

- Loaded resources are displayed.
- Unselected resources are also shown.
- Vehicle capacity is never exceeded.

Algorithm Used

0/1 Knapsack Dynamic Programming

Formula:

$DP[i][w]$

=

max

$(DP[i-1][w],$

$DP[i-1][w-w_i] + v_i)$

Purpose

Maximize utility while respecting cargo limits.

DEMO 5: Branch and Bound Fleet Assignment

Objective

Demonstrate optimal vehicle-to-zone assignment.

Steps

1. Open **Fleet Assignment**.
2. Review the **Branch and Bound Decision Panel**.
3. Open **Simulation Analytics**.

Explanation

The system evaluates multiple vehicle assignment combinations.

Each assignment cost includes:

- Travel Cost
- Route Risk
- Cargo Penalty

Cost Function

Total Cost

=

Travel Cost

-

Risk Cost

-

Unserved Resource Penalty

Benefit

Suboptimal assignments are discarded early through pruning.

This reduces unnecessary computations.

DEMO 6: Real-Time Hazard Recalculation

Objective

Demonstrate dynamic route recalculation under changing crisis conditions.

Steps

1. Open **Route Visualizer**.
2. Select **Flood Emergency**.
3. Inspect route to Zone I (Burbank).
4. Change scenario to **Conflict Zone**.
5. Inspect the route again.

Observation

The route changes immediately.

Explanation

Different scenarios modify road conditions.

Dijkstra recalculates paths using updated edge weights.

Benefit

Real-time adaptability during emergencies.

DEMO 7: Mathematical Verification of Priority Scores

Objective

Validate the heap priority calculation.

Steps

Submit:

Zone A (Glendale)

Severity = Critical

Population = 12,000

Observed Priority Score:

31.2

Submit:

Zone E (Santa Monica)

Severity = Low

Population = 10,000

Observed Priority Score:

1.0

Verification

Priority Score

=

(Severity Weight $\times$ 10)

•

(Population $\div$ 10,000)

For Glendale:

(3 $\times$ 10)

•

(12,000 $\div$ 10,000)

=

30 + 1.2

=

31.2

For Santa Monica:

(0×10)

•

$(10,000 \div 10,000)$

=

$0 + 1$

=

1.0

This confirms that heap ranking is mathematically deterministic.

DEMO 8: Database Resilience & Failover Architecture

Objective

Demonstrate system reliability during database failures.

Architecture

Primary Database:

Supabase PostgreSQL

Backup Database:

SQLite

Steps

1. Observe the green **Supabase Mode** indicator.
2. Simulate a database failure.
3. The system automatically switches to SQLite.

Observation

The status indicator changes to:

SQLite Mode

Explanation

The application catches connection failures and redirects operations to the local database.

Result

All algorithms continue functioning:

- Heap Operations
- Resource Allocation
- Dijkstra Routing
- Knapsack Optimization
- Fleet Assignment

without service interruption.

Final Demonstration Summary

The Crisis Logistics AI Optimizer integrates five major DAA algorithms:

1. Max-Heap Priority Queue
 - Emergency request prioritization
2. Greedy Allocation
 - Resource distribution
3. Dijkstra Algorithm
 - Safe route optimization
4. 0/1 Knapsack
 - Vehicle cargo optimization
5. Branch and Bound
 - Fleet assignment optimization

Together, these algorithms create an intelligent logistics platform capable of delivering resources efficiently during disasters, emergencies, and crisis situations.